Requirements Analysis

November 2021

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1 Introduction

1.1 Purpose of the system

In a network of inter-communicating industrial components, a large amount of data is transferred in between these components. It is required to process the packets send over this network in order to use the processed data for further computation. For this purpose a preprocessor is used. Visualizing results of the preprocessor helps with gaining an overview of the network data and the steps of preprocessor.

Preprocessor The preprocessor takes a list of packets (PCAP) as input and returns a coordinate point for each packet. There are two methods of preprocessing used in this system: Principle Component Analysis (PCA) and using an autoencoder. For both methods the user can configure the scaling method and the normalization method.

PCAP **p**acket **cap**ture - File containing data about network packets. Each packet uses at least one networking protocol. A protocol determines the format of the packet and how the data is transmitted in a network. In the context of this system and in the rest of this document the term PCAP is used for a file containing information about the captured packets of the network.

PCA **p**rinciple **c**omponent **a**nalysis - PCA is used to reduce the dimensionality of data.

Autoencoder An autoencoder is a neural network. It is used to learn codings for each packet. Using this system the user can configure the autoencoder by setting the following: the number of hidden layers, the number of nodes in those layers, the loss function and the number of epochs for the training. The hidden layers are the layers between the input and output layer of the neural network. A loss function is a function that the autoencoder tries to minimize in the learning process. The number of epochs is the number of times the neural network trains on the whole PCAP.

Goals and Criteria of the system By using this system the user should be able to judge the results and performance of different preprocessing configurations. This is achieved by allowing the configuration of the preprocessor to be done in a graphical user interface which also displays a visual representations of the PCAP and the results and performance of the preprocessing. The configurations can be given a name and can be saved to a file.
The visualization of the PCAP takes the form of a network topology graph where components are represented by nodes and connections by edges. If the user hovers over a node with the mouse pointer, the MAC address of the device is shown.

The results of the preprocessing are displayed in a two dimensional coordinate system with points representing the network packets while the interpretation and the range of the x-Axis and y-Axis are determined by the preprocessor. In order to provide a better understanding of the results the packets are shaped and color coded based on the highest protocol used. If the user hovers over a point in the coordinate system the following information about the packet is displayed: sender and receiver IP addresses (if the protocol uses IP), sender and receiver MAC addresses, the networking protocol used and a timestamp of the network packet. Both the network visualization and the results visualization can be filtered by networking protocols.
The performance is visualized based on the preprocessing method. For the Autoencoder the performance graph consists of the training accuracy, the validation accuracy and the test accuracy. If PCA is selected, the performance is described by two values: the loss on the train data and the loss on the test data. To improve understanding of the results, the user can zoom in or out of any of the three graphs described above.
Another important functionality for this system is the comparison of different preprocessor configurations with each other. To make this possible, the user can display results and performance of configurations next to each other or merge the graphs into one.
To share results, the user can save graphs as images individually or save an session to be reopened at another time or at another device.

Limitations This system can not be used to analyse a network in real time, instead only one PCAP file can be analysed and that PCAP file does not change while running the program. If the user changes the configuration of the preprocessor the visual representations do not update unless a button to run the preprocessing is clicked. The PCAP can not be filtered by time or protocols. Only the visualizations can be filtered by protocols.

1.2 Scope of the system

This program is used in the context of industrial networks. In order to achieve better results using a machine learning model, the network packets of the industrial network are preprocessed. The purpose of this program is to help with this step by visualizing the preprocessing results which allows optimization of the preprocessing configuration.

1.2.1 Product environment

This program runs inside a modern web browser. It is intended to run in the current stable versions of Firefox (94.0.2) or Chromium-based browsers like Google Chrome (96.0.4664.45) or Microsoft Edge (96.0.1054.34).

2 Proposed system

2.1 Overview

/F10/	Load a PCAP file into the program
/F20/	Configuration of the preprocessor
/F22/	Run the preprocessor
/F30/	Visual representation of the network topology
/F40/	Visual representation of the preprocessor results
/F50/	Performance analysis of the preprocessor
/F60/	Zoom functionality for graphs
/F70/	Save the graphs and configurations
/F80/	Comparison of preprocessor performance and results of different runs
/F90/	Asynchronous interaction with panels

2.1.1 User interface

Since the goal of this system is understanding preprocessing configurations through visualization of data, the graphical user interface (GUI) is the main focus. The components of the GUI are explained in the following paragraphs.

The main window of the program: This window is the main focus of the system, it is where the user can configure the preprocessor, run it and see the results and performance of the preprocessing.

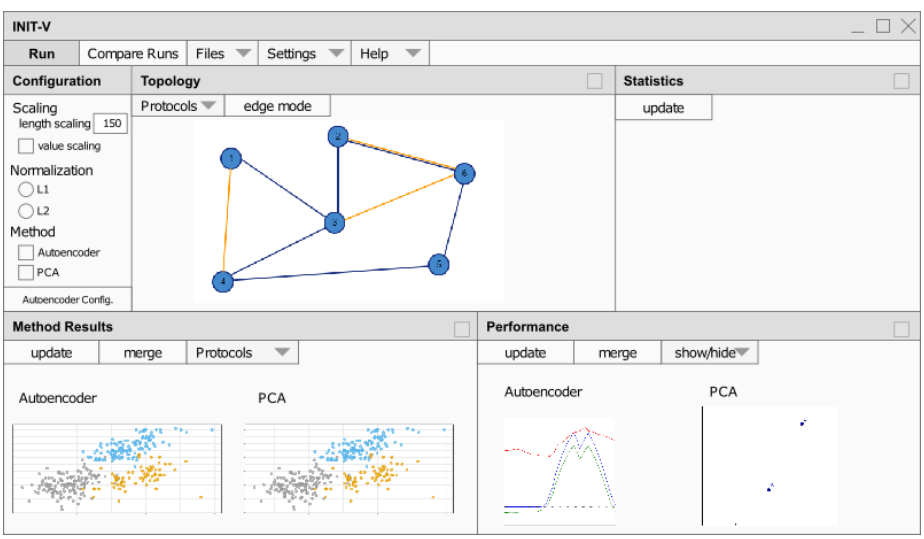

Figure 1: The main window of the program

Menu Bar: The main window consists of a menu bar with two buttons and three menus:

- The "Run" button to run the preprocessor with the configuration set in the configuration panel.
- The "Compare Runs" button to open a window where two runs can be compared side by side.
- A files menu for handling file input and output.
- A settings menu to manage configurations.
- A help menu.

Menus: There are three menus in the menu bar at the top of the window. The files menu, the settings menu and the help menu.

Files menu: The files menu consists of four menu items:

- The "Open" menu item is used to either open a PCAP file or to open a previously saved session.
- The "Save" menu item is used to save the current session to a file. If the current session has not been saved before, it has the same effect as the "Save As..." menu item.
- The "Save As..." menu item opens a dialog where the current session can be saved as a file. The file can be given a name and the location in the file system can be selected.
- The "Export As..." menu item creates a PDF document which includes configurations, results and performances of the current sessions (Could-have functionality).

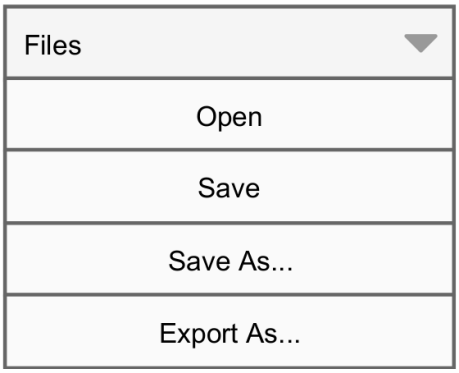

Figure 2: The files menu

Settings menu: The settings menu consists of five menu items:

- The "Default Configuration" menu item sets all values of the preprocessor configuration to their default values.
- The "Load Configuration" menu item lets the user open a previously saved configuration file.
- The "Save Configuration" menu item lets the user name a preprocessor configuration and adds it to the list of configurations at the bottom of the menu.
- The "Export Configuration" menu item opens a dialog where the current configuration of the preprocessor can be saved as a file. The file can be given a name and the location in the file system can be selected.
- At the bottom of the menu is a list of saved configurations. By selecting one of the configurations, its values are set in the configuration panel.

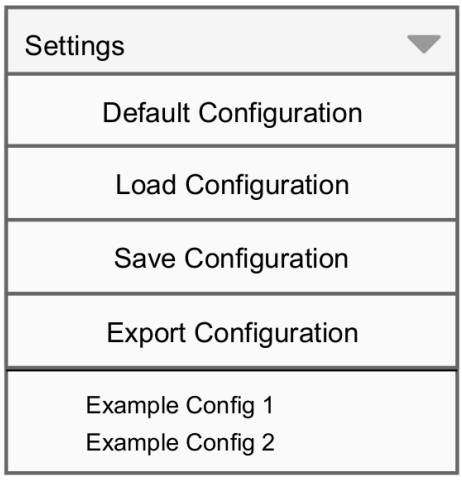

Figure 3: The settings menu

Help menu: The help menu consists of an "About" menu item. This menu item opens a window displaying the name, logo and version number of the program.

Figure 4: The help menu

Configuration panel: A panel to change the preprocessor configuration is located at the top left of the main window, here the user can set the length scaling value and check whether to use value scaling. It also includes the selection of the normalization method and of the preprocessing method (autoencoder or PCA). At the bottom of the panel is a button which opens a window where the details of the autoencoder configuration can be set.

Autoencoder configuration: This window allows the user to:

- Input the number of hidden layers and the number of nodes in the hidden layers of the neural network.
- Choose the loss function between MSE and MAE.
- Input the number of epochs for the training of the neural network.
- Select the optimizer from a drop down menu.

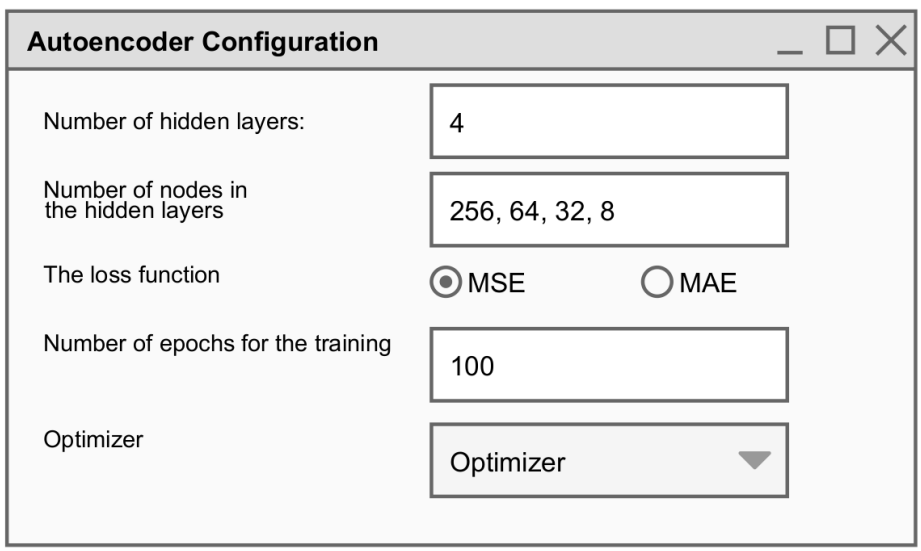

Figure 5: The autoencoder configuration window

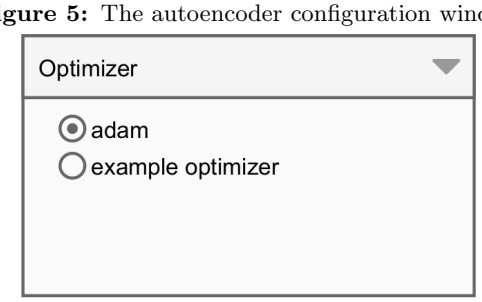

Figure 6: The optimizer drop down menu

Topology panel: A panel showing the network topology of the PCAP is located at the top middle of the main window. The topology of the network is a graph where components are represented by nodes and connections by edges. A drop down menu to select protocols is located at the top left of the panel. Edges will be drawn according to the selected protocols. Next to that is a button to switch between drawing edges for connections and drawing edges for each protocol. If the user hovers over a node with the mouse pointer the MAC address of the device is shown.

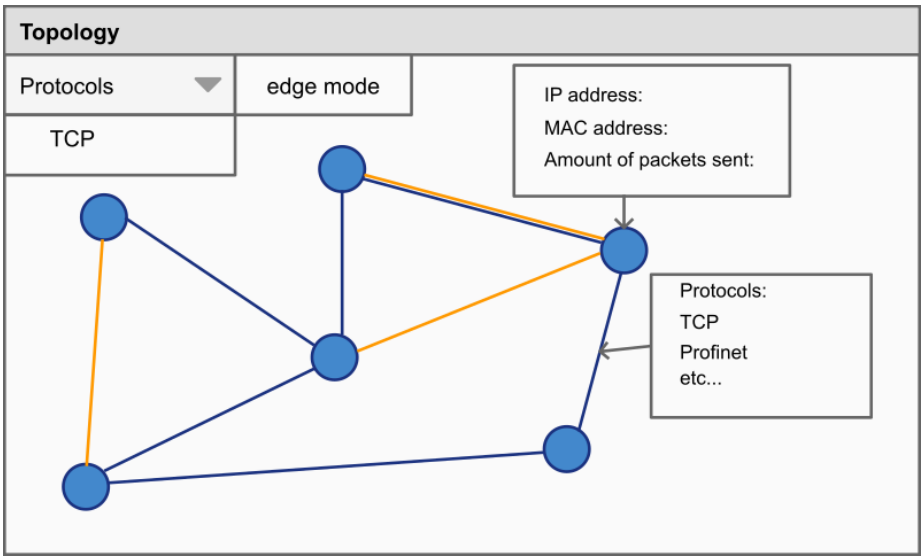

Figure 7: Detailed view of the topology panel

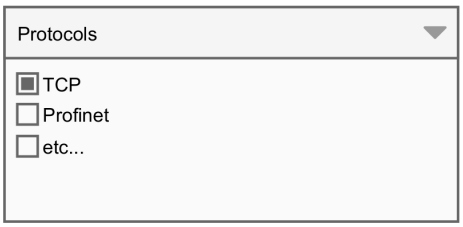

Figure 8: The protocols menu for the topology panel

Statistics panel: A panel displaying statistics about the results and performance of the preprocessor is located at the top right of the main window. There is a button to update the statistics located at the top left of the panel.

Results panel: A panel showing the results of the preprocessing is located at the bottom left of the main window.

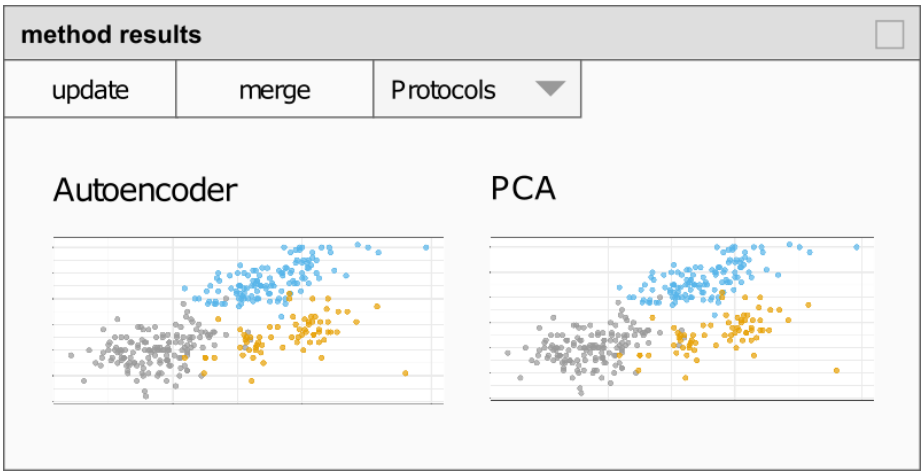

Figure 9: The method results window

There are three buttons on this panel:

- An update button to only update the results panel.
- A merge button to merge the results from the PCA and autoencoder preprocessing methods into one graph.
- A drop down menu to select protocols. Only packets of the selected protocols will be drawn on the results graphs.

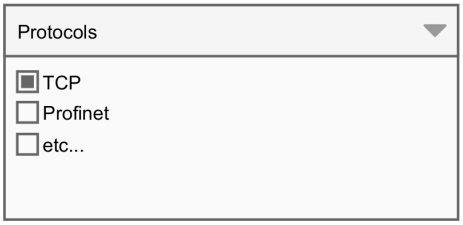

Figure 10: The protocols menu for the results panel

Performance panel: A panel displaying the performance of the preprocessor is located at the bottom right of the main window. The performance includes the training, validation and test accuracy of the autoencoder and the loss on the train data and the loss on the test data of the PCA.

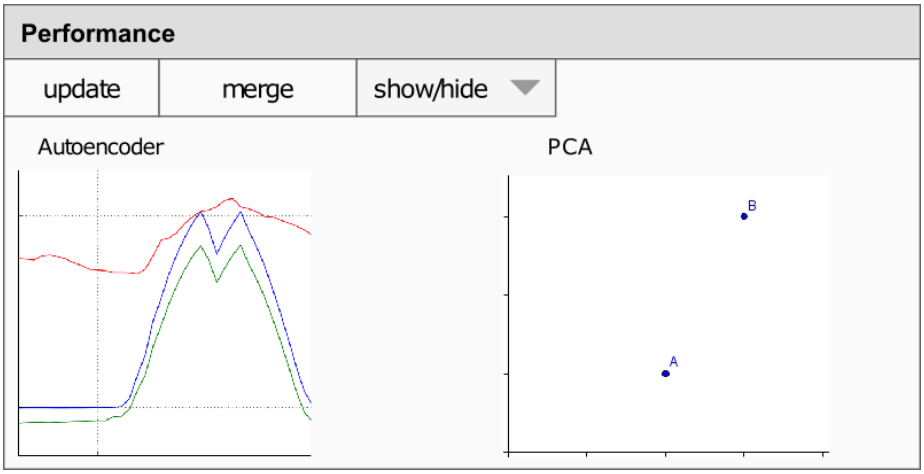

Figure 11: The performance window

There are three buttons on this panel:

- An update button to only update the performance panel.
- A merge button to merge the results from PCA and autoencoder into one graph.
- A button to open a window where the user can decide which values and lines to show or hide.

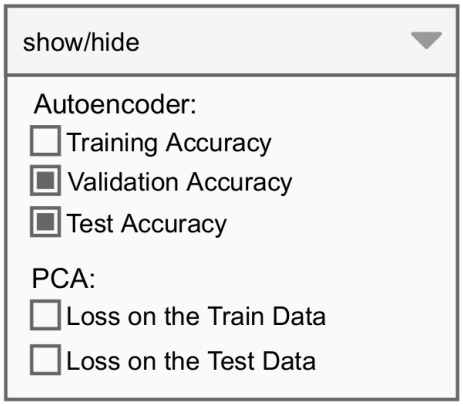

Figure 12: The window where the user can select which lines and values to show or hide on the performance panel

Comparison window: If the user clicks on the "Compare Runs" button in the menu bar, the comparison window opens. At the top left of that window is a drop down menu where the user can select the runs to compare. Next to it is a button, which merges the results and performances of the compared graphs into one graph.

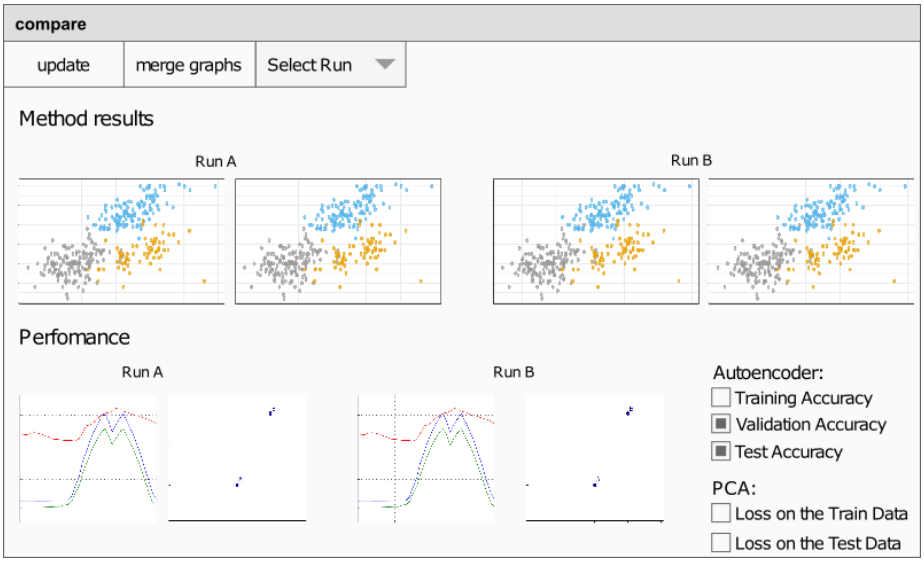

Figure 13: The comparison window

Program launcher: When starting the program, a launcher window appears with the option to either open a PCAP file or to reopen the last session of the program on this device or to open any saved session.

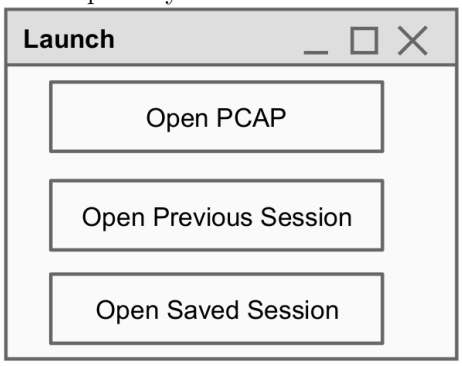

Figure 14: The window appearing when launching the program

2.2 Functional requirements

2.2.1 Must-haves

/F10/ Loading a PCAP file into the program
There a multiple ways of loading a PCAP:

/F11/ Loading a PCAP file directly
Precondition: -none-
Effect: The PCAP is loaded and the visual representation of the network appears (/F30/).
Description:

- The user can open a PCAP by selecting a button in the File menu.
- The user can choose between:
  - The PCAP replaces the currently used PCAP.
  - A new instance of the program opens with the PCAP loaded.

/F12/ Opening the previous session
Precondition: The program has previously been used on the computer.
Effect: The previous session is restored.
Description:

- When the program is launched the user can choose to open the previous session.
- Upon choosing that option the previous session is loaded:
  - The last used configuration of the preprocessor is selected.
  - The visual representation of the network appears (/F30/).
  - The results of the last preprocessing appear (/F40/).
  - The performance graphs of the last preprocessing appear.

/F13/ Opening a saved session
Precondition: During a past session a save file has been created.
Effect: The state of the program when the save file was created is restored.
Description:

- The user can open a saved session by selecting a button in the File menu.
- Upon choosing that option the saved session is loaded:
  - The last used configuration of the preprocessor is selected.
  - The visual representation of the network appears (/F30/).
  - The results of the last preprocessing appear (/F40/).
  - The performance graphs of the last preprocessing appear.

/F20/ Configuration of the preprocessor
Precondition: -none-
Effect: The preprocessor will use the configuration.
Description:

- Changing the preprocessor configuration is possible by setting the following:
  - Set a value for the length scaling.
  - Select whether to use value scaling.
  - Select between None, L1 and L2 normalization.
  - Select from Autoencoder and PCA.
  - Configure the neural network of the Autoencoder:
    1. Number of hidden layers
    2. Number of nodes in the hidden layers
    3. The loss function: MSE or MAE
    4. Number of epochs for the training
    5. Optimizer
- Preprocessing configurations can be named and saved.
- The default values of required configurations can be changed.

/F21/ Default values for the preprocessor configuration
Precondition: -none-
Effects: The preprocessor is configured with default values, allowing it to be run.
Description: The default values are:

- Length scaling: 150
- Value scaling: Off
- Normalization: None
- Autoencoder and PCA are selected.
- Number of hidden layers: 4
- Number of nodes in hidden layers: (256, 64, 32, 8)
- Loss function: MSE
- Number of epochs for the training: 100
- Optimizer: adam

/F21a/ Change default values of the preprocessor
Precondition: -none-
Effects: The preprocessor is configured with the changed default values.
Description:

- After clicking on "Default Configuration" in the "Settings" menu, a window open allowing the default configuration to be changed.

/F22/ Run the preprocessor
Precondition: A PCAP was loaded and the preprocessor has been configured.
Effect: The results and performance graphs appear.
Description: The configured preprocessor is run on the loaded PCAP by clicking the "Run" button in the menu bar.

/F30/ Visual representation of the network topology
Precondition: A PCAP was loaded.
Effect: Visual representation of the network topology appears in a panel.
Description:

- Representation of the network topology as an undirected graph.
- Nodes represent devices and edges can be configured to show all communication between two devices as a single edge or have an edge for each protocol.
- Each protocol has a distinct color for the edge. If an edge represents all communication between two devices, then its color is the color of the protocol that was most used in that communication.
- Through a checkbox menu be able to show the topology only of the devices taking part in communication using chosen protocols.
- Show the MAC and IP address of a device in a popup box if the mouse pointer hovers on the node representing that device.
- If edges represent all communication between two devices show all protocols in a popup box if the mouse pointer hovers on an edge.

/F31/ Configure /F30/ edge properties
Precondition: A PCAP was loaded.
Effect: Configure edges on the network topology panel (/F30/) to represent either all communication between two devices or to only represent communication on one protocol.
Description: As described in /F30/, the edges can either represent all communication between two devices or communication using a single protocol. The user can configure this to their needs. In a topology that has communication on multiple protocols, choosing the edge for all communication simplifies the graph, making it more understandable.

/F32/ Device details appear in a popup box when the mouse pointer hovers over a node in the network topology graph
Precondition: A PCAP was loaded, the mouse pointer hovers over a node in the network topology graph.
Effects: A popup box appears at the note containing information about the device represented by that node.
Description: The popup box consists of following information:

- IP address of the device
- MAC address of the device

/F40/ Visual representation of the preprocessor results
Precondition: A PCAP was loaded, the preprocessor has been configured and the preprocessor has been run by clicking the "Run Preprocessing" button.
Effect: Visual representation of the network topology appears in a panel.
Description:

- Have two graphs dedicated to showing a visual representation of the result of the preprocessing step. One graph for the Autoencoder results, the other for the PCA results.
- The representation is a two dimensional graph of the network packets. The parameters and the range of the x- and y-axis are chosen by the preprocessor.
- Each point on the graph represents a network packet. How the point is drawn is determined by the highest protocol used by the packet. Every protocol has a distinct visual representation.
- Show sender IP, receiver IP(if IP is used by the protocol), sender MAC address, receiver MAC address, protocol and timestamp of a packet if the mouse pointer is hovering over the point that represents that packet (/F41/).
- Only show selected protocols on the result graphs (/F42/).

/F41/ Show additional information about a packet when hovering over it in /F40/
Precondition: A PCAP was loaded, the preprocessor has been configured and the preprocessor has been run by clicking the "Run Preprocessing" button, the mouse pointer hovers on a point on the graph /F40/.
Effects: A popup box appears on the point that consists of information about the packet represented by that point.
Description: The popup box consists of the following information:

- Sender IP address (if used by the protocol)
- Receiver IP address (if used by the protocol)
- Sender MAC address
- Receiver MAC address
- Protocol of the packet
- Timestamp of the packet

/F42/ Show only points representing packets of selected protocols in /F40/
Precondition: A PCAP was loaded, the preprocessor has been configured and the preprocessor has been run by clicking the "Run Preprocessing" button.
Effects: Points representing selected protocols by the user are shown on the graph /F40/.
Description: The user can select specific protocols to reduce the points shown on the graph /F40/ by selecting the protocols in a drop down menu.

/F50/ Performance analysis of the preprocessor
Precondition: A PCAP was loaded, the preprocessor has been configured and the preprocessor has been run by clicking a button.
Effect: The performance analysis appears in a panel.
Description:

- The performance analysis is shown as two 2d graphs one for the Autoencoder on for the PCA.
- The two graphs can be merged into one.
- The performance of the Autoencoder is described by three lines:
  - Training Accuracy
  - Validation Accuracy
  - Test Accuracy
- The performance of the PCA is described by two values:
  - Loss on the train data
  - Loss on the test data
- The lines and values are colored differently and can be hidden or shown using a check box selection.

/F60/ Zoom functionality for graphs
Precondition: The visual representations of the network topology, results or performance are shown.
Effect: Increased or decreased viewed area in a graph or changed the position of the view.
Description:

- Zooming in and out of the visual representations of the network topology, results and performance analysis.
- When zoomed in, moving the view of the graphs.
- When zoomed in or out, display a button to reset the view to the default position.

/F70/ Save the graphs and configurations
There are multiple ways of saving results and configurations:

/F71/ Save the visual representations of the network topology, results and performance analysis as images
Precondition: The graphs which the user wants to save are displayed in their panels.
Effect: An image is created and can be saved as a file on the user's device.
Description:

- After right clicking a graph a menu appears with the option to save the graph as an image file.
- The image depicts the selected graph as it currently appears in the panel (zoom level and view position).

/F72/ Save the current state of the program as a file
Precondition: A PCAP has been loaded.
Effect: A file containing the current state of the program is created and can be saved on the user's device.
Description:

- After clicking the "Save" or "Save As..." menu button in the File menu a file is created containing the current state of the program.
- The file can be saved to be reopened at another time or at another device.

/F72a/ Display a reminder when closing the program and there are unsaved changes
Precondition: There are unsaved changes made in the session.
Effect: Display a reminder that there are unsaved changes and provide an option to save these changes.

/F73/ Save a configuration as a file
Precondition: The preprocessor has been configured.
Effect: A file containing the configuration is created and can be saved on the user's device.

/F80/ Comparison of preprocessor performance and results of different runs
Precondition: The preprocessor has been run at least two times.
Effects: By overlaying graphs a comparison of preprocessor runs becomes possible.
Description:

- The comparison is displayed in a window which opens when the user clicks "Compare Runs" in the menu bar.
- The user can select from a list of runs, with various preprocessing configurations done on this session, by using a drop down menu.
- Selected runs results and performance analysis can be compared by overlaying the graphs or by looking at two runs results and performance analysis side by side.

/F90/ Asynchronous interaction with panels
Precondition: -none-
Effect: While the preprocessor is running the program remains responsive to user input.
Description:

- The visualizations of the network topology can be interacted with(as defined in /F30/ and /F60/) while the preprocessor is running.
- The visualizations of the performance analysis can be interacted with(as defined in /F60/) while the preprocessor is running.
- The results of the last preprocessor run can be interacted with(as defined in /F40/ and /F60/) while the preprocessor is running.

2.2.2 Should-haves

/F100/ Have subwindow(s) dedicated to statistics of the input PCAP file
Precondition: A PCAP file has been loaded.
Effects: The subwindow shows various statistics of the PCAP file.
Description: The subwindow shows various statistics specified by the following functional requirements /F102/-/F103/.

/F101/ Change the shown statistic on /F100/
Precondition: -none-
Effects: Changes the shown statistic on the subwindow defined in /F100/.
Description: There are various available statistic modes defined in /F102/-/F103/. The user can change the statistic shown on this window.

/F102/ Show a bar graph of the protocols used on the PCAP file on the subwindow defined in /F100/
Precondition: A PCAP file has been loaded and the bar graph option was selected by the user via /F101/.
Effects: The window defined in /F100/ shows a bar graph of the protocols of the packets sent in the PCAP file.
Description: Each bar on the bar graph represents the sum of the packets sent with a protocol.

/F103/ Show miscellaneous statistics of the PCAP file on the subwindow defined in /F100/
Precondition: A PCAP file has been loaded and the miscellaneous statistics option was selected by the user via /F101/.
Effects: The window defined in /F100/ shows a table of the protocols of miscellaneous statistics of the PCAP file.
Description: By "miscellaneous statistics" following statistics of the PCAP file are shown:

- Count of devices
- Count of used protocols
- Count of packets
- Device that sent the most packets

2.2.3 Could-haves

/F33/ Communication details in popup box when mouse pointer hovers on an edge in /F30/
Precondition: A PCAP was loaded, each edge is configured to represent all communication between two devices, the mouse pointer hovers on an edge in /F30/.
Effects: A popup box appears on the edge containing information on the communication between the two devices' nodes connected by the hovered on edge
Description: The popup box consists of following information:

- Protocols used between the two devices
- Amount of exchanged packets

/F200/ Graphs can be opened in their own window
Precondition: The preprocessor was run.
Effects: A window displaying only the graph is opened.
Description: After clicking a button next to the graph, the graph is opened in another window.

/F210/ Commonly used functionality can be used with key bindings
Precondition: -none-
Effects: By using the key combination specified, the associated action gets executed.
Description:

- Key bindings for:
  - Opening a saved session (/F13/): Ctrl + O
  - Saving a session (/F72/): Ctrl + S
  - Saving a session by selecting a file (/F72/): Ctrl + Alt + S
  - Run the preprocessor: Ctrl + R
- These key bindings can be changed by the user in a settings menu.
- Changed key binds are saved between sessions.
- In the settings menu there is a button to restore the key bindings to the default bindings.

/F220/ Include identifiers inside of a saved session (/F72/)
Precondition: -none-
Effect: The save file contains identifiers.
Description: The identifiers are:

- Hash to identify program version.
- Hash to identify the PCAP.
- Hash to identify the results.

/F230/ If the program is handling large amounts of data, display a warning
Precondition: A PCAP file containing information on more than 2 million packets has been selected to be opened.
Effect: A popup appears telling the user that the program could be less responsive.

/F240/ Create a report of the current session and save it as a PDF file
Precondition: At least one configuration was run.
Effect: A PDF document containing information about the session is saved.
Description: The document includes:

- The configurations of the preprocessing runs.
- The results of those configurations.
- The performance graphs of those configurations.

2.3 Nonfunctional requirements

2.3.1 Must-haves

/P10/ Showing results from preprocessing done in this session should be responsive
Precondition: At least two preprocessing configurations have been run.
Effects: The results of the older preprocessing configuration is shown in under 1 second.

/S10/ The program is free from data leaks
Description:

- No information about the PCAP should be accessible by inspecting the source of the website.

/E10/ Adding functionality to the program later on is possible without restructuring the program
Description: The developers of this project (the authors of this document) will not be maintaining the project after March 2022, therefore the design of the software should be adequately structured to allow expansion of it. Following are features that must be accordingly designed for easy expandability:

- Adding or removing another menu to a menu bar.
- Adding or removing menu items to a menu.
- Adding or removing options to configure the preprocessor.
- Adding or removing statistics.
- Sub-window management.

/E11/ The implementation and the design is thoroughly documented
Description: In order to make understanding and extending the program possible, the code and the design of the program is documented in detail.

2.4 Product data

2.4.1 PCAP

The user loads a PCAP file to the software, it is locally processed.

2.4.2 Saved session

A file can be created to store the state of the program. This file can be saved or opened.

2.4.3 Configuration file

If /F210/ is implemented a configuration file is used to store key bindings on the users machine.

2.4.4 Image files

All graphs can be saved as PNG images to the device of the user.

2.5 Development environment

2.5.1 IDE

PyCharm 2021 will be used as an IDE.

2.5.2 Frameworks

- **plotly** will be used for the GUI.
- **pytest** will be used for testing.

2.5.3 Version control

git will be used as version control.

2.5.4 Documentation

LaTeX will be used for documentation.

3 Testing

3.1 Test cases

Distinction The goal of this section is establishing test cases to be able validate the system defined by this document. Therefore we will define these in a similar fashion as the functional requirements. The basic test cases will cover just the core functionality of the system and advanced cases will target the more abstract uses this software should provide.

Basic Cases:

/T10/ Starting the system.

/T20/ Launch options will show after startup.

- /T21/ Load previous session on start.
- /T22/ Load PCAP on start.
- /T23/ Open a saved session on start.

/T30/ Load PCAP.

/T40/ Load session.

/T50/ Save session.

/T60/ Export results.

/T70/ Update Panels.

- /T71/ Topology
- /T72/ Statistics
- /T73/ Method Results
- /T74/ Performance

/T80/ Change default settings.

/T90/ Load a configuration

/T100/ Save current configuration.

/T110/ Export current configuration.

/T120/ Select current configuration from saved ones.

/T130/ Select protocols to be shown in the Topology.

/T140/ Input a value for length scaling.

/T150/ Select value scaling.

/T160/ Select normalisation method between L1 or L2.

/T170/ Select PCA and/or Autoencoder

/T180/ Change the Autoencoder configuration.

- /T181/ Set number of hidden layers.
- /T182/ Set number of nodes in hidden layers.
- /T183/ Select between "MSE" and "MAE" for the loss function.
- /T184/ Set number of epochs for the training.
- /T185/ Select optimizer.

/T190/ Run the preprocessor.

/T200/ Select which graphs to show in the performance panel.

/T210/ Exit program.

- /T211/ Save on exit.
- /T212/ Exit without saving.

/T220/ Open the compare window.

/T230/ Save graphs as PNG images.

- /T231/ Topology.
- /T232/ Method results.
- /T233/ Performance.
- /T234/ Statistics.
- /T235/ Performance comparison.
- /T236/ Method results comparison.

/T240/ Select a run in the compare window.

Advanced Cases:

/T310/ Show info on node in topology when hovering over it.

/T320/ Show info of edge in topology when hovering over it.

/T340/ Merge the Performance graphs.

/T350/ Merge the Method results graphs.

/T360/ Select which results to include while exporting.

/T370/ Show graphs in a separate windows.

- /T371/ Topology.

/T372/ Method results.

/T373/ Performance.

/T374/ Statistics.

/T375/ Performance comparison.

/T376/ Method results comparison.

/T380/ Close separate windows of graphs.

/T381/ Topology.

/T382/ Method results.

/T383/ Performance.

/T384/ Statistics.

/T385/ Performance comparison.

/T386/ Method results comparison.

3.2 Test scenarios

3.2.1 Scenario 1 - Comparing two configurations

The user starts the program, chooses to load a PCAP, turns on value scaling, saves the configuration as "vsconfig" and then runs the preprocessor. Afterwards the user configures the autoencoder by changing the loss function to "MAE" and changing the number of training epochs to 50. The preprocessor is run and then the user selects "Compare Runs". In the compare window the user selects the "vsconfig" run and saves the performance comparison graphs as a PNG image.

- /T10/ Starting the system.
- /T22/ Load PCAP on start.
- /T150/ Select value scaling.
- /T100/ Save current configuration.
- /T190/ Run the preprocessor.
- /T183/ Select between "MSE" and "MAE" for the loss function.
- /T184/ Set number of epochs for the training.
- /T190/ Run the preprocessor.
- /T220/ Open the compare window.
- /T240/ Select a run in the compare window.
- /T235/ Save graphs as PNG images (Performance comparison).

3.2.2 Scenario 2 - Loading previous session

The user starts the program, chooses to load a PCAP then runs the preprocessor, exits the program and saves in the exit dialogue. They then start the program and load the previous session.

- /T10/ Starting the system.
- /T22/ Load PCAP on start.
- /T190/ Run the preprocessor.
- /T211/ Save on exit.
- /T10/ Starting the system.
- /T21/ Load previous session on start.

3.2.3 Scenario 3 - Export as PNG

The user starts the program, chooses to load a PCAP, runs the preprocessor and exports every graph as PNG. Afterwards they change the number of hidden layers, rerun and export the method results and performance each as PNG and exit the program without saving.

- /T10/ Starting the system.
- /T22/ Load PCAP on start.
- /T190/ Run the preprocessor.
- /T230/ Save graphs as PNG images.
- /T181/ Set number of hidden layers
- /T190/ Run the preprocessor.
- /T232/ Save graph as PNG image: Method results.
- /T233/ Save graph as PNG image: Performance.
- /T212/ Exit without saving.

3.2.4 Scenario 4 - Opening all graphs in separate windows

The user starts the program, chooses to load a PCAP, runs the preprocessor and opens every graph in a separate window: first topology, then method results, then performance and then statistics. They then close the windows in that same order and exit without saving.

- /T10/ Starting the system.
- /T22/ Load PCAP on start.

- /T190/ Run the preprocessor.
- /T371/ Show graph in a separate window: Topology.
- /T372/ Show graph in a separate window: Method results.
- /T373/ Show graph in a separate window: Performance.
- /T374/ Show graph in a separate window: Statistics.
- /T381/ Show graph in a separate window: Topology.
- /T382/ Show graph in a separate window: Method results.
- /T383/ Show graph in a separate window: Performance.
- /T384/ Show graph in a separate window: Statistics.
- /T212/ Exit without saving.

Glossary

autoencoder An autoencoder is a neural network. It can be used to learn codings for each packet. One of the preproccessing methods. 3, 28, 31

IP **I**nternet **P**rotocol - Standard protocol to address other computers. 3

MAC **M**edia-**A**ccess-**C**ontrol - MAC addresses are used to uniquely identify hardware devices in a network. 2, 3

network topology The structure of a network describing how the components are connected to each other, presented by a graph, where the nodes are components, and the edges are connections. 4, 8, 14, 18

PCA **p**rinciple **c**omponent **a**nalysis - One of the preproccessing methods. PCA can be used to reduce the dimensionality of a more dimensional data. 2, 3, 31

PCAP **p**acket **cap**ture - File containing data of Network Packets. 1, 2, 3, 12, 20, 24, 28, 29, 31

preprocessor The preprocessor takes a list of packets (PCAP) sent over the network as input and returns a coordinate point for each packet. Also see PCA and autoencoder . 2, 3, 4, 13, 14, 15, 17, 19, 28, 29, 31

protocol A system of communication rules and formats, by which multiple components of the network transmit data. 3, 16, 20

session One or multiple runs with one or multiple preprocessor configurations on a single PCAP. 1, 12, 22, 23, 24